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# Fórmula de la varianza

**Proposición 1 (Fórmula de la varianza).** *Sea* $X : \Omega \longrightarrow \mathbb{R} \in \mathcal{L}^1(\mathbb{P})$ *v.a.*

$$\implies \mathbb{V}(X) = \mathbb{E}\left[X^2\right] - \left(\mathbb{E}\left[X\right]\right)^2.$$

***Demostración*:** Sea $X \in \mathcal{L}^1(\mathbb{P})$ una variable aleatoria, entonces

$$\begin{aligned}\mathbb{V}(X) &= \mathbb{E}\left[|X - \mathbb{E}[X]|^2\right] = \mathbb{E}\left[X^2 - 2X\mathbb{E}[X] + (\mathbb{E}[X])^2\right] \\ &\overset{\substack{\text{lineal} \\ \downarrow}}{=} \mathbb{E}\left[X^2\right] - 2\left(\mathbb{E}[X]\right)^2 + \left(\mathbb{E}[X]\right)^2 = \mathbb{E}\left[X^2\right] - \left(\mathbb{E}[X]\right)^2.\end{aligned}$$

■

## Referenciado en

- `prop-varianza-sum-var-aleatorias-indep`

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